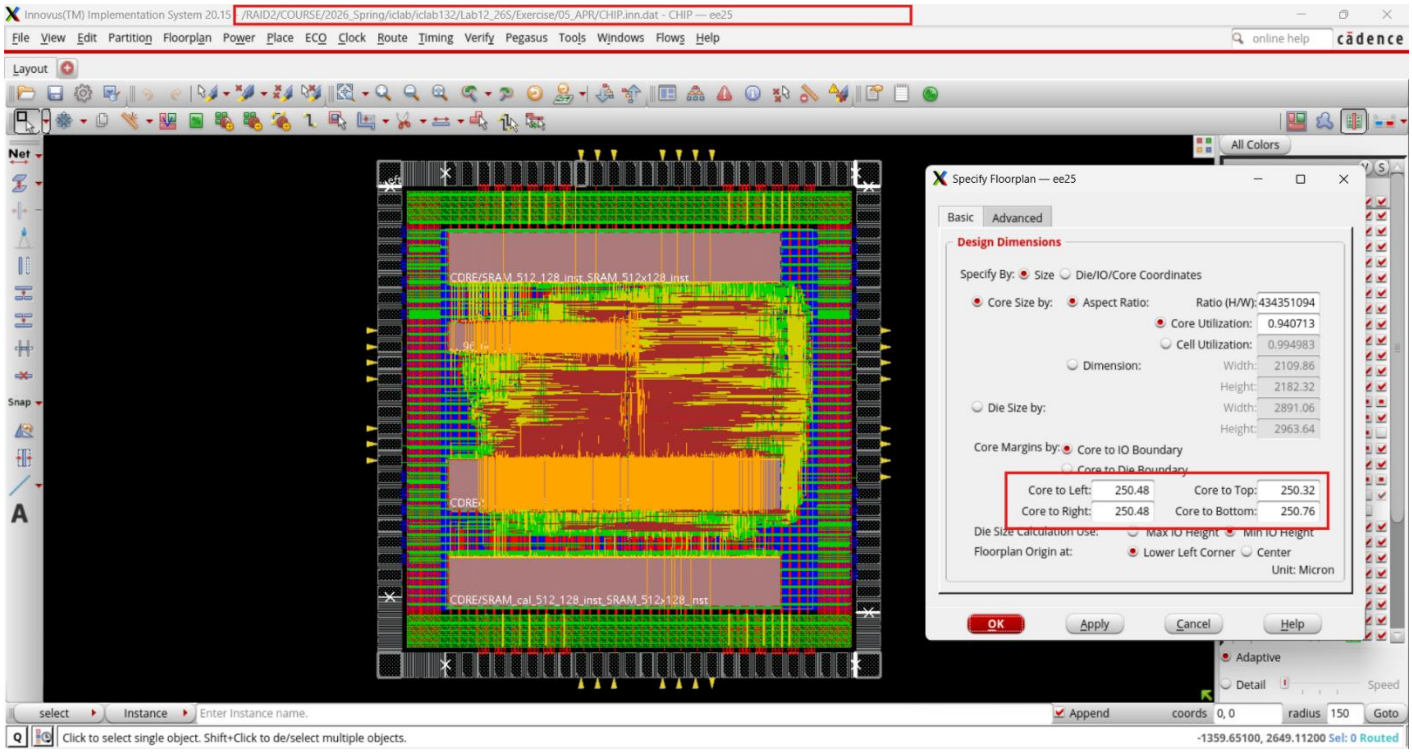
Report

iclab132 蘇峻霆

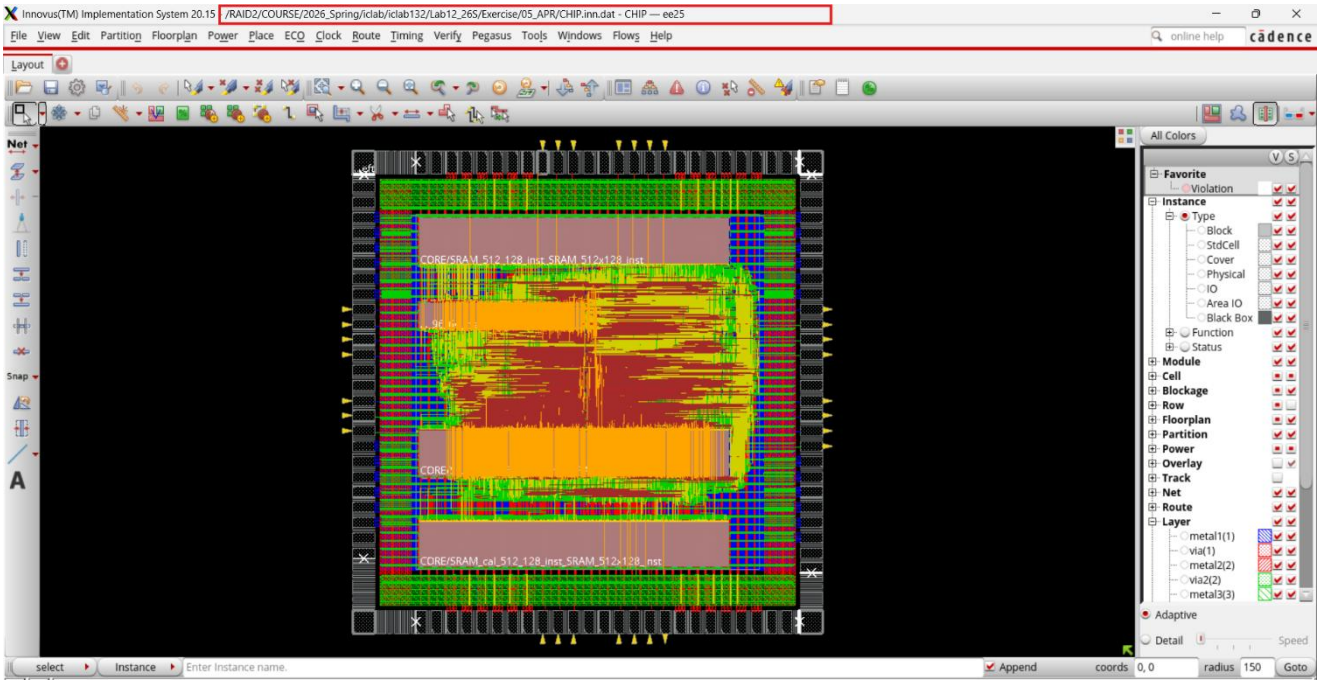
1. Core to IO boundary :



- **Core to IO boundary: 250**

2. Core Ring :

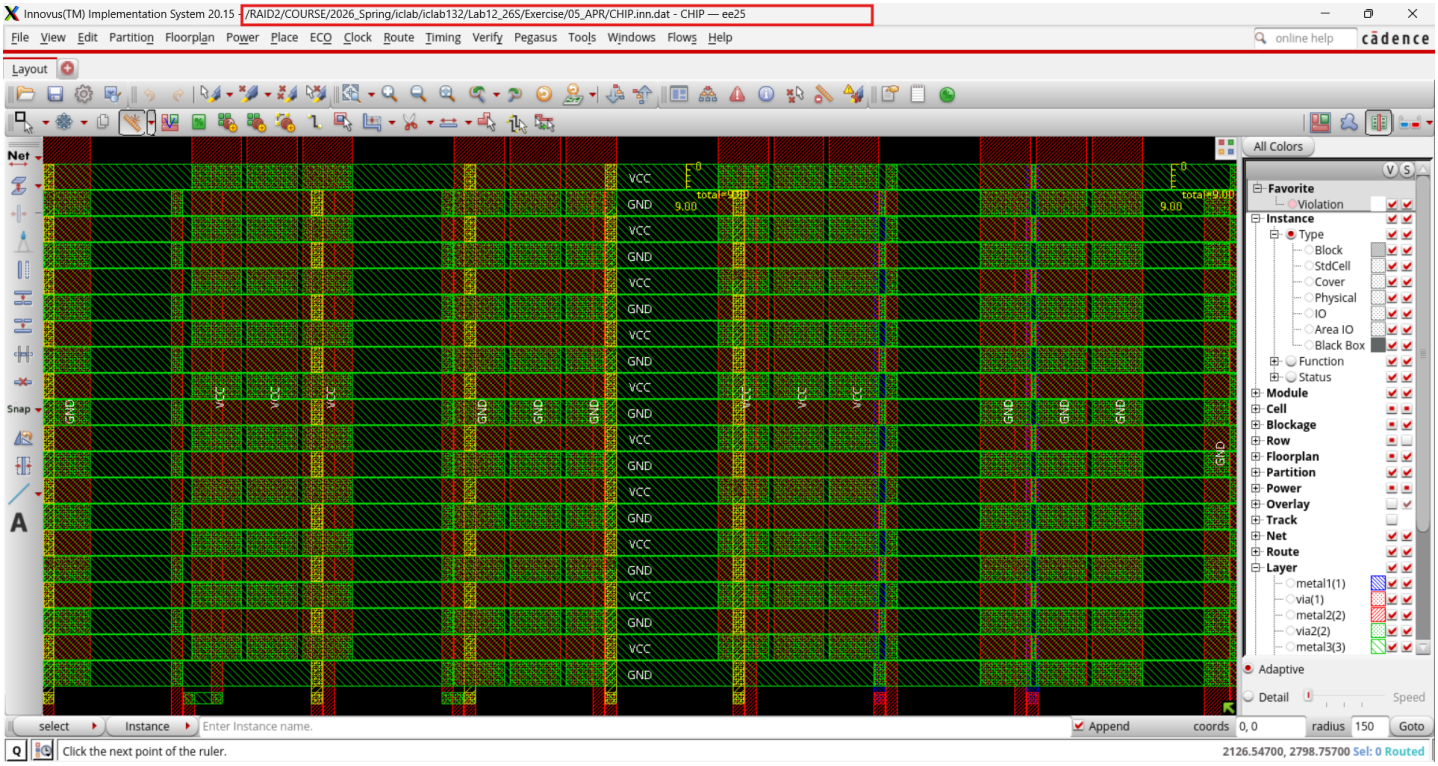
- **layout**



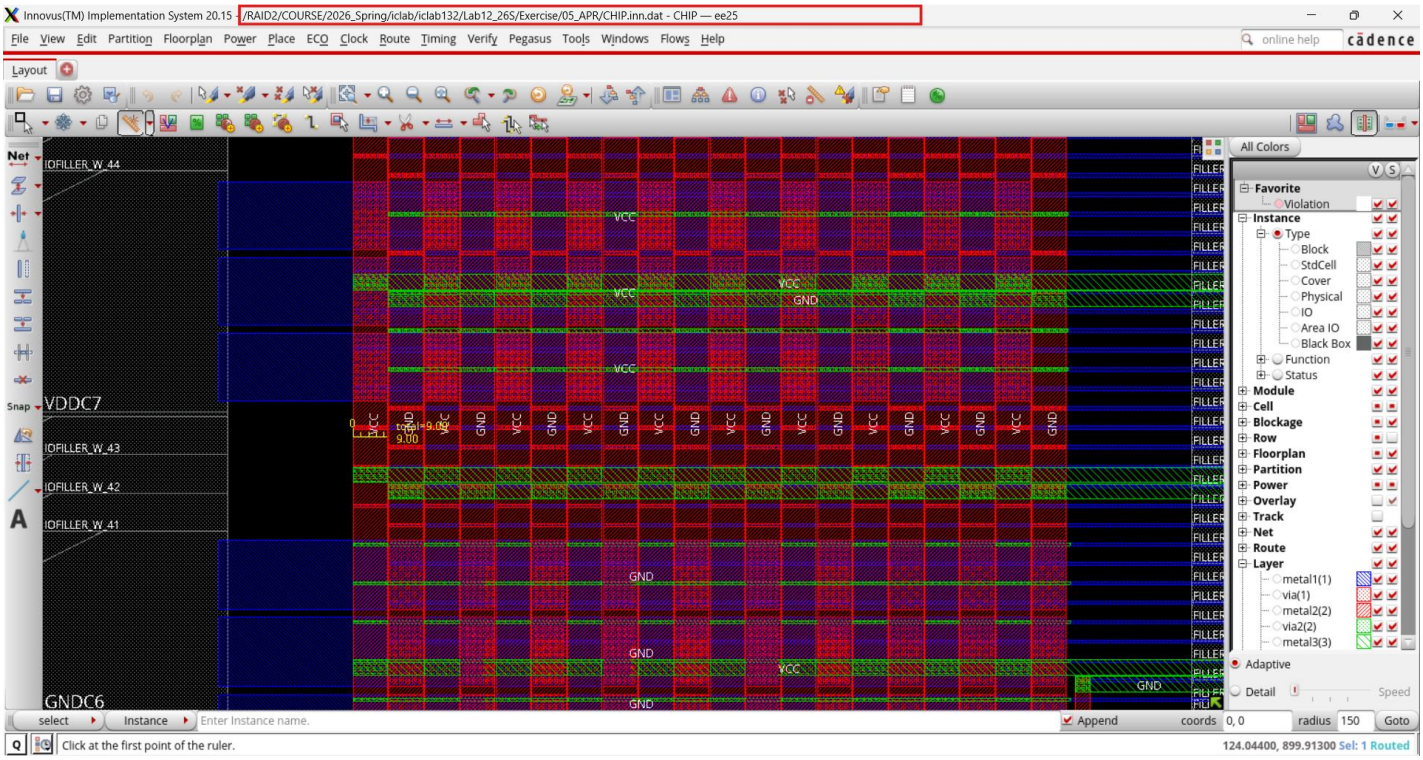


- **wire group**
- **interleaving**
- **10 pairs**

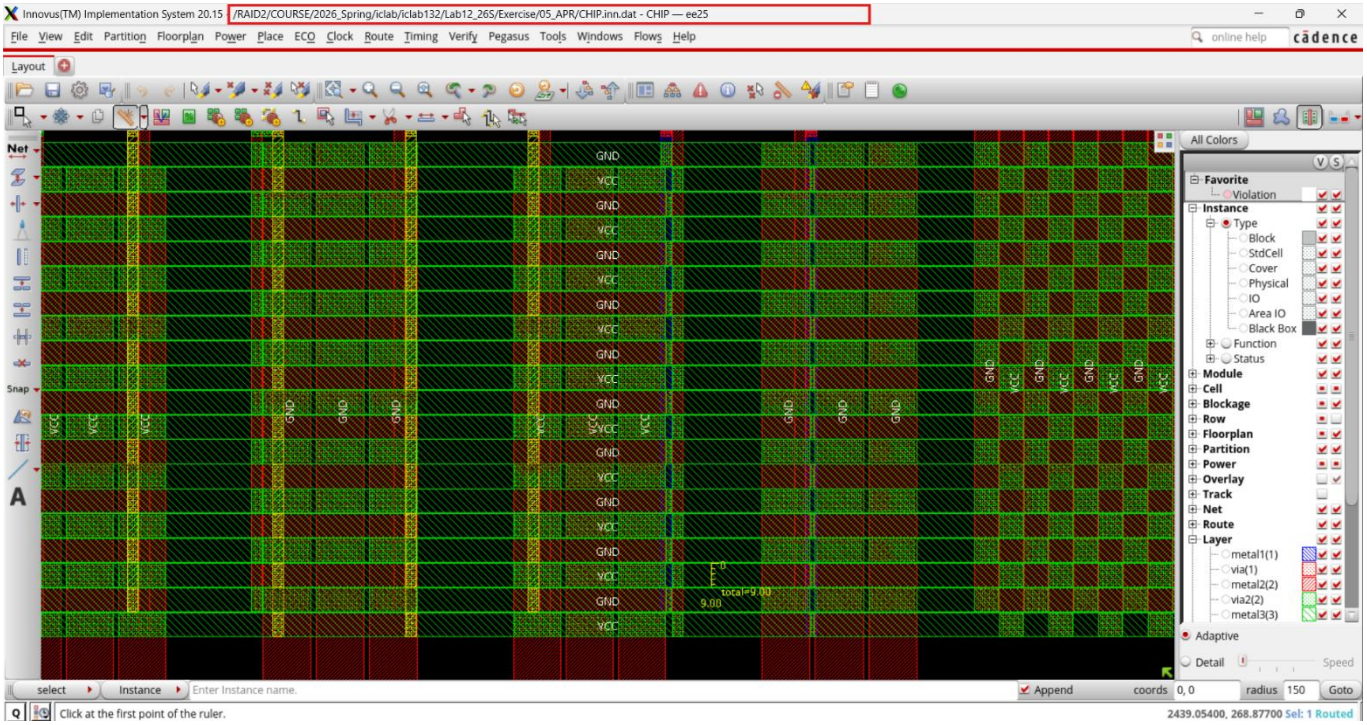
  - **top**



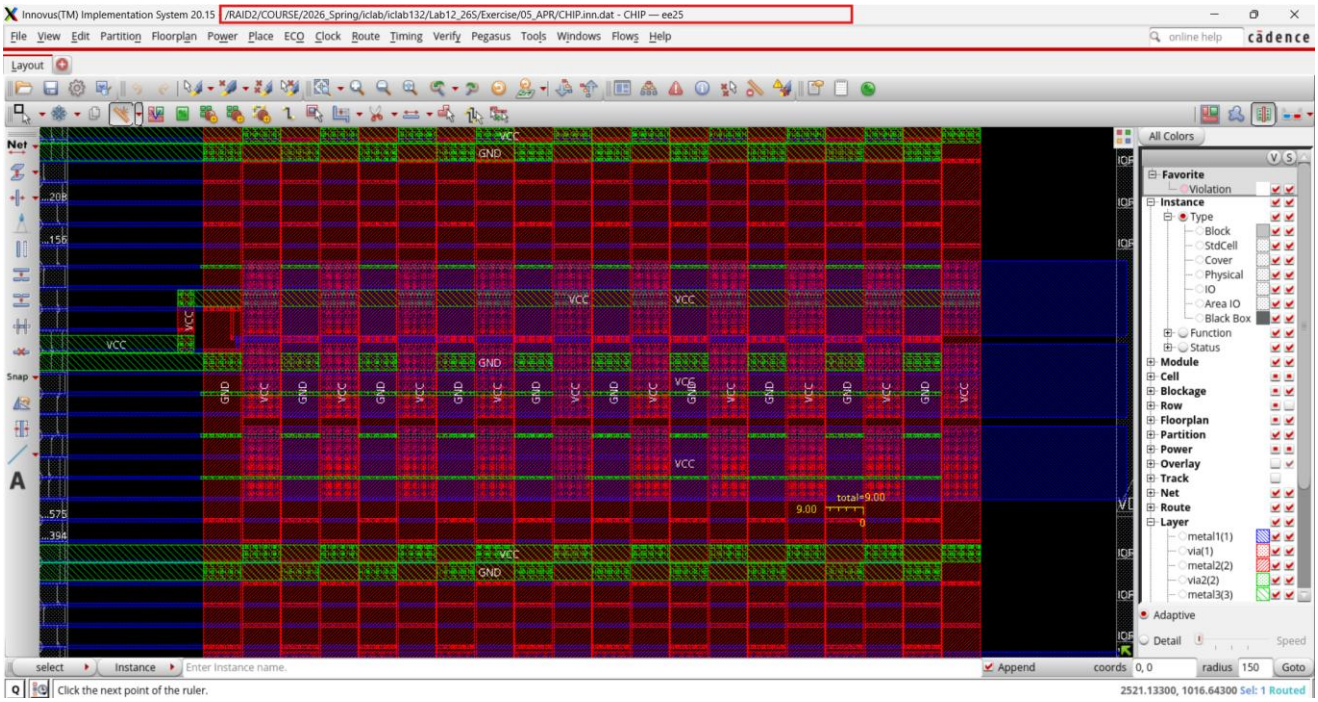
- **left**


Innovus(TM) Implementation System 20.15 - /RAID2/COURSE/2026_Spring/iclab/iclab132/Lab12_26S/Exercise/05_APR/CHIP.inn.dat - CHIP — ee25
File View Edit Partition Floorplan Power Place ECO Clock Route Timing Verify Pegasus Tools Windows Flows Help
cadence
Layout
VDDC7
GNDC6
IOFILLER_W_44
IOFILLER_W_43
IOFILLER_W_42
IOFILLER_W_41
All Colors
Favorite
Violation
Instance
Type
Block
StdCell
Cover
Physical
IO
Area IO
Black Box
Function
Status
Module
Cell
Blockage
Row
Floorplan
Partition
Power
Overlay
Track
Net
Route
Layer
metal1(1)
via(1)
metal2(2)
via2(2)
metal3(3)
Adaptive
Detail
Speed
select
Instance
Enter Instance name.
Append
coords 0, 0
radius 150
Goto
Click at the first point of the ruler.
124.04400, 899.91300 Sel: 1 Routed

- **bottom**

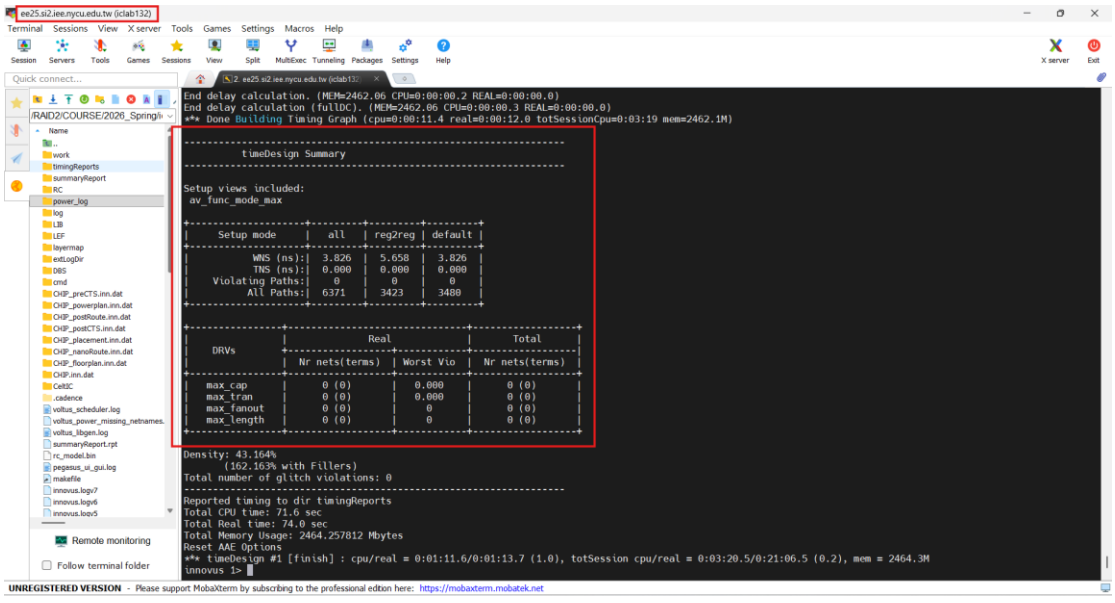

Innovus(TM) Implementation System 20.15 - /RAID2/COURSE/2026_Spring/iclab/iclab132/Lab12_26S/Exercise/05_APR/CHIP.inn.dat - CHIP — ee25
File View Edit Partition Floorplan Power Place ECO Clock Route Timing Verify Pegasus Tools Windows Flows Help
cadence
Layout
All Colors
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Violation
Instance
Type
Block
StdCell
Cover
Physical
IO
Area IO
Black Box
Function
Status
Module
Cell
Blockage
Row
Floorplan
Partition
Power
Overlay
Track
Net
Route
Layer
metal1(1)
via(1)
metal2(2)
via2(2)
metal3(3)
Adaptive
Detail
Speed
select
Instance
Enter Instance name.
Append
coords 0, 0
radius 150
Goto
Click at the first point of the ruler.
2439.05400, 268.87700 Sel: 1 Routed

- **right**



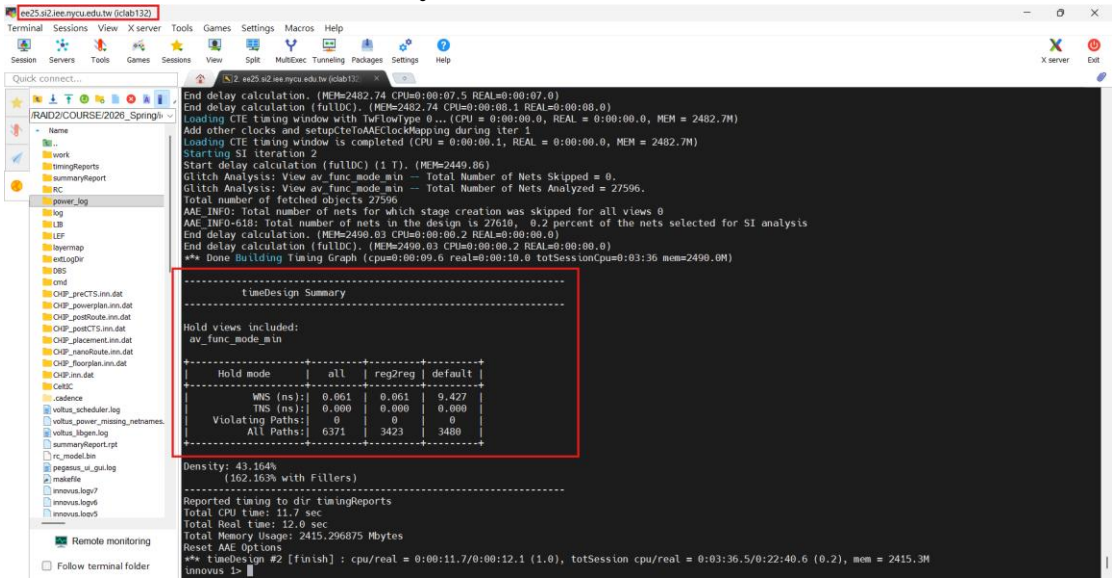
- **Top & Bottom: metal layer must be odd and width is fixed to 9.**
- **Left & Right: metal layer must be even and width is fixed to 9.**

3. Post-Route setup time analysis :



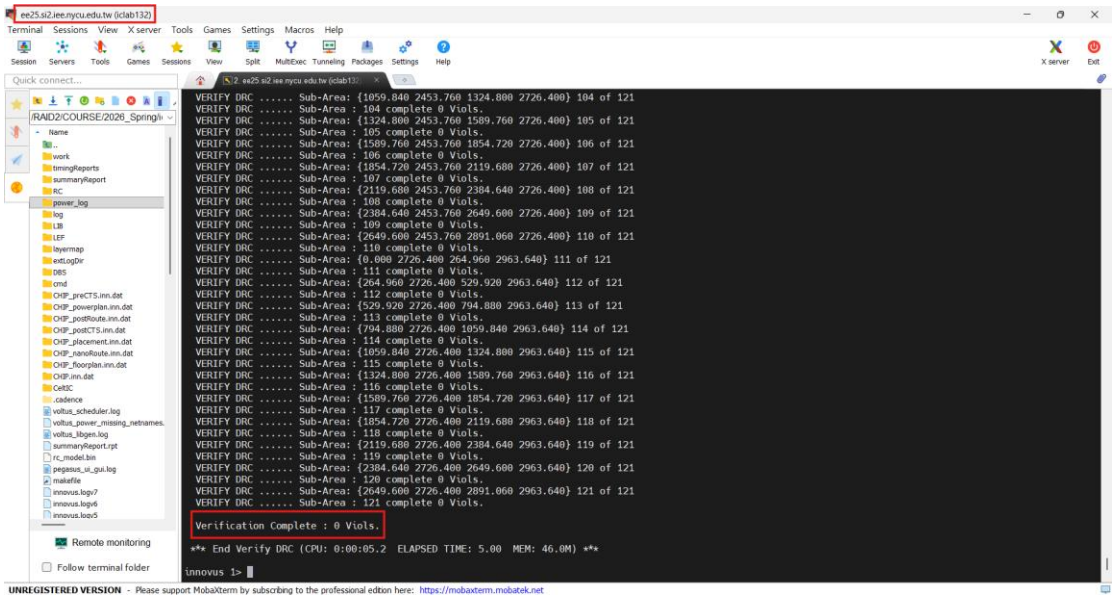
- **No negative slack**
- **Zero DRVs**

4. Post-Route hold time analysis :



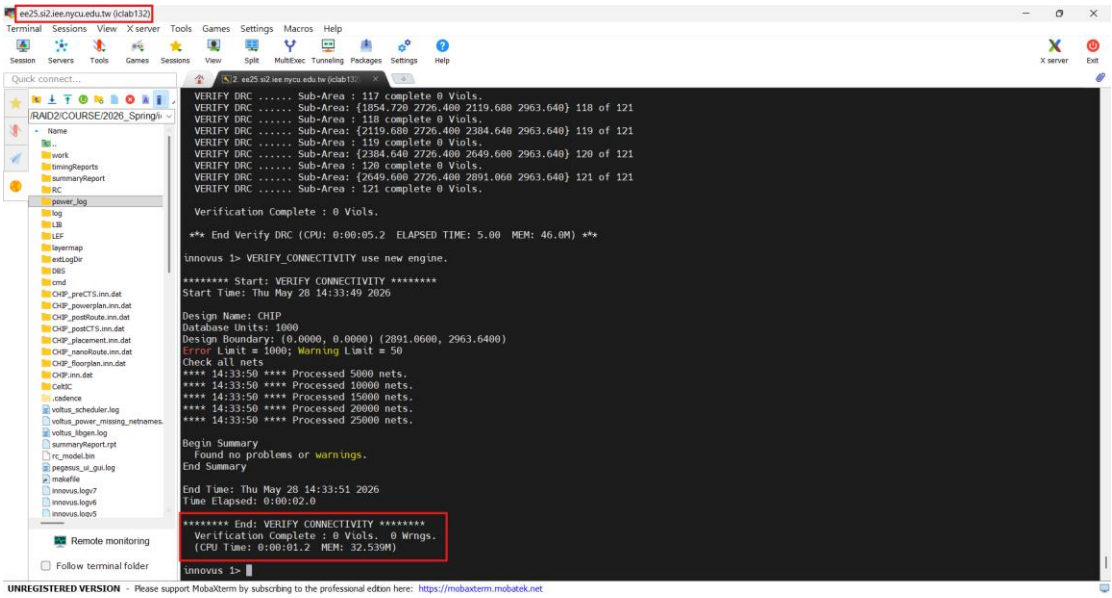
- **No negative slack**

5. DRC result :



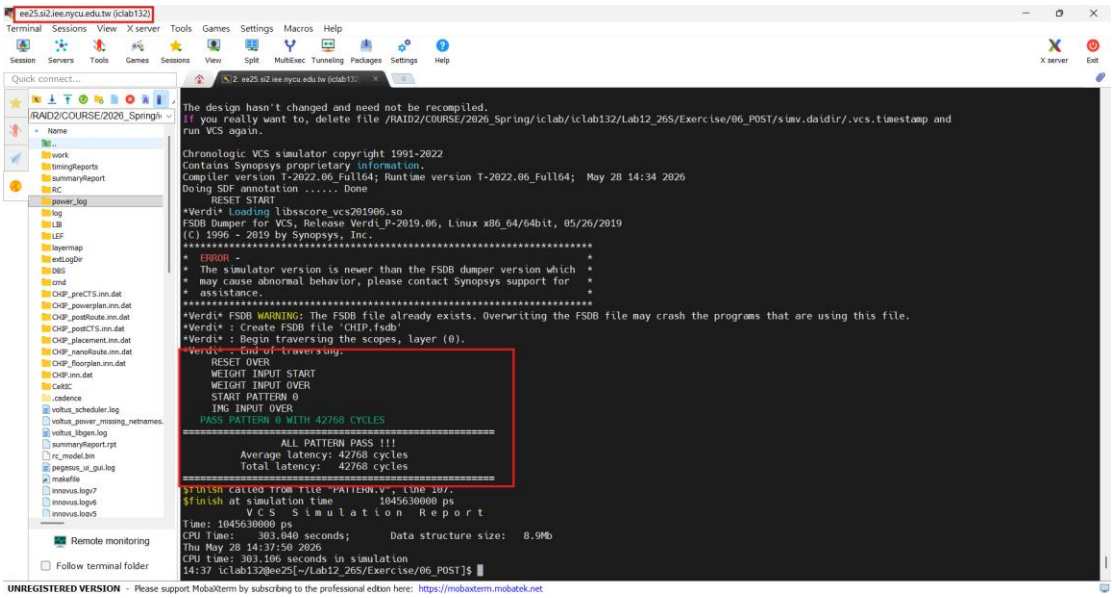
- **Zero violation**

6. LVS result :



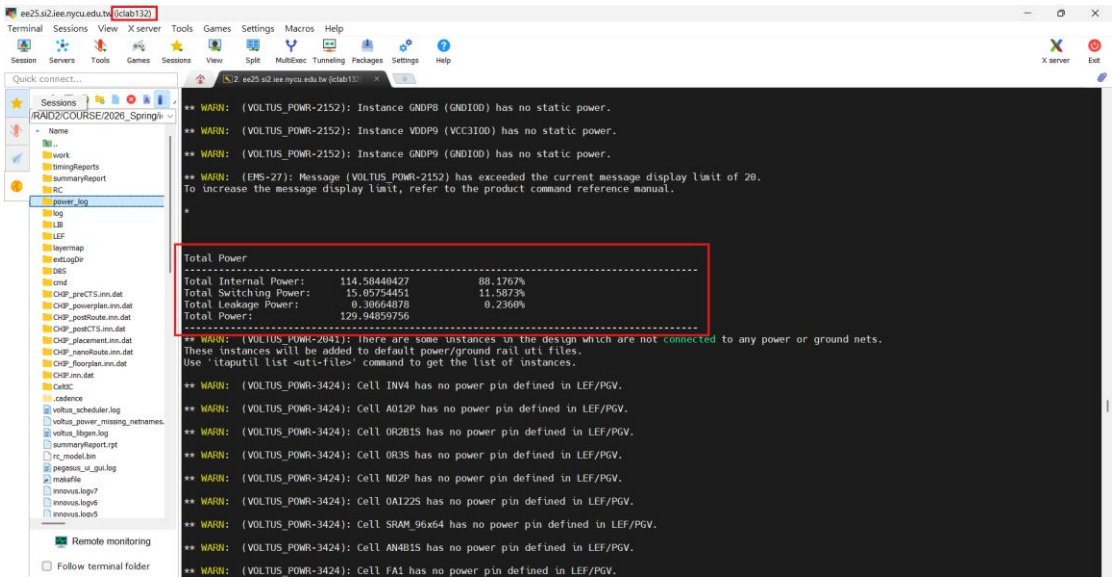
- **Zero violation**

7. Post Layout simulation result :

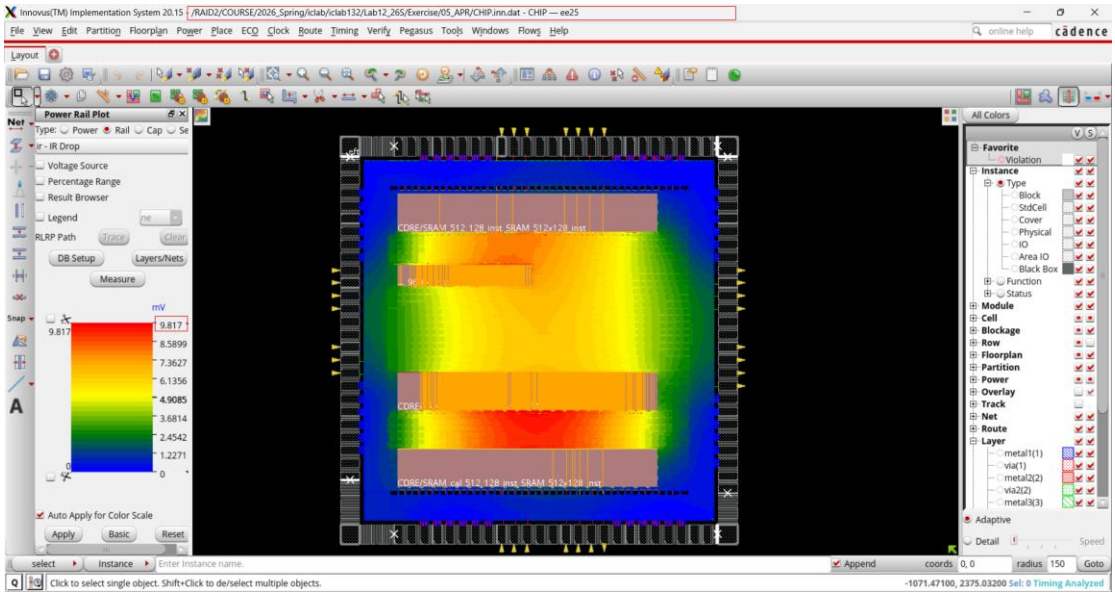


- **Latency is the same as gate level simulation**

8. Power result :



9. IR Drop Results :



■ **Max IR Drop: 9.817mV**

1. 我多開core power pad，開了24組，並且分散擺放讓每個邊都有6組，而且讓VDD和GND相連，這樣可以減少那一條power的path。
2. cycle time增加到20 ns，讓power消耗變小，使core power pad供應充足。
3. power stripes的密集度增加，從間隔100變成間隔50，大幅度供應各區域的power。如果SRAM之間的空間沒有塞入power stripe，IR drop會飆升，把SRAM分散擺放也可以舒緩這個問題。
4. core ratio設0.55，因為要增加power stripe密集度，所以standard cell會擺放更分散，需要有大空間讓他擺放，防止wire阻塞。